Computational PDEs: Classical Approaches and Physics-Informed Learning

Day 4 — Schottky–Klein primes, close circles, and PINNs

Hiroyuki Miyoshi (The University of Tokyo)

12 November 2026

The University of Tokyo
ELTE Erasmus+ ICM Teaching Mobility
Faculty of Informatics, Eötvös Loránd University

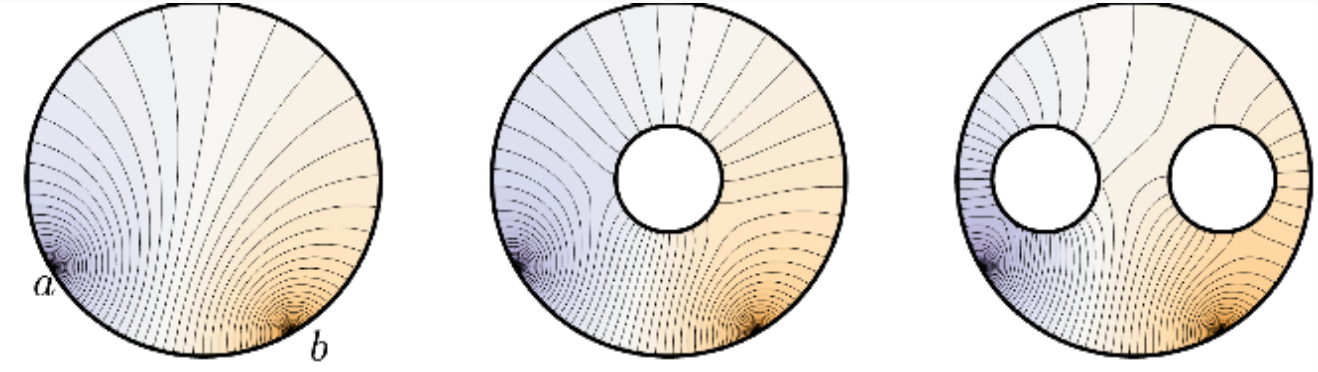
Outline

1. Annulus: the product $P(\zeta, \rho)$ and the vortex
2. Several holes: Schottky generators and the prime function ω
3. Close circles: truncation becomes expensive
4. A PINN from the same identities
5. Application: two holes set the symmetry of a cell alignment

Annulus and the product P

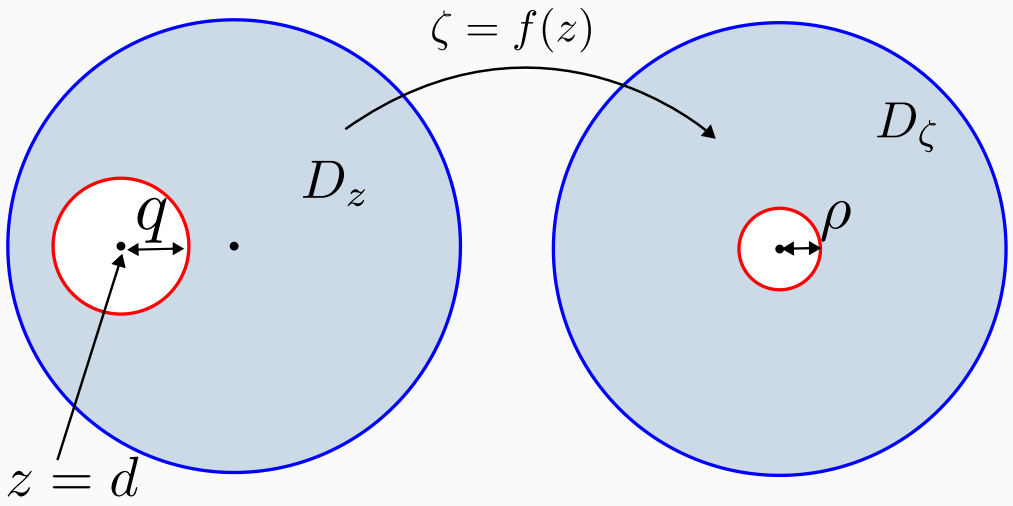
Wednesday in one picture

- Simply connected $\to$ unit disk (Tuesday).
- One hole $\to$ **annulus** $D_\zeta = \{\rho < |\zeta| < 1\}$.
- Two or more holes $\to$ unit disk with circular holes (circular canonical domain).



The annulus and infinite images

- Point singularity at $\zeta = a$: demand $\psi = \text{const}$ on both C_0 ($|\zeta| = 1$) and C_1 ($|\zeta| = \rho$).
- One reflection fixes one circle and breaks the other.
- The procedure never stops $\rightarrow$ an **infinite product**.



The function $P(\zeta, \rho)$

$$P(\zeta, \rho) := (1 - \zeta) \prod_{n=1}^{\infty} (1 - \rho^{2n}\zeta)\,(1 - \rho^{2n}\zeta^{-1}). \tag{1}$$

Two identities

$$P(\zeta^{-1}, \rho) = -\zeta^{-1}\, P(\zeta, \rho) \qquad \text{(inversion)},$$
$$P(\rho^2\zeta, \rho) = -\zeta^{-1}\, P(\zeta, \rho) \qquad \text{(quasi-periodicity)}.$$

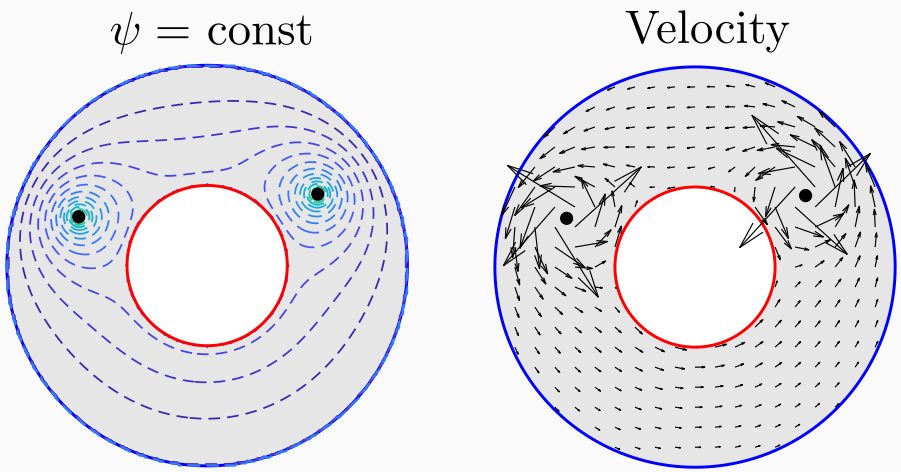
Unique zero in the annulus at $\zeta = 1$. Truncate the product at $N_{\max}$ for numerics. Those two relations become the PINN loss later this morning.

Point vortex from P

On the annulus,

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} \log \left(\frac{|a|\, P(\zeta/a, \rho)}{P(\zeta\, \overline{a}, \rho)} \right). \tag{2}$$

- The identities of P make $\operatorname{Im} w$ constant on C_0 and C_1.
- Wednesday: build P by the product.
- The same formula will hold with ω once there are several holes.



Several holes

From the annulus to several holes

Wednesday built P by inverting in two circles. One more hole is another circle, not another factor ρ^{2n}.

The signed images on the annulus were

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} \Big(\log(\zeta - a) - \log(\zeta - 1/\overline{a}) - \log(\zeta - \rho^2/\overline{a}) + \log(\zeta - \rho^2 a) + \log(\zeta - a/\rho^2) - \cdots \Big).$$

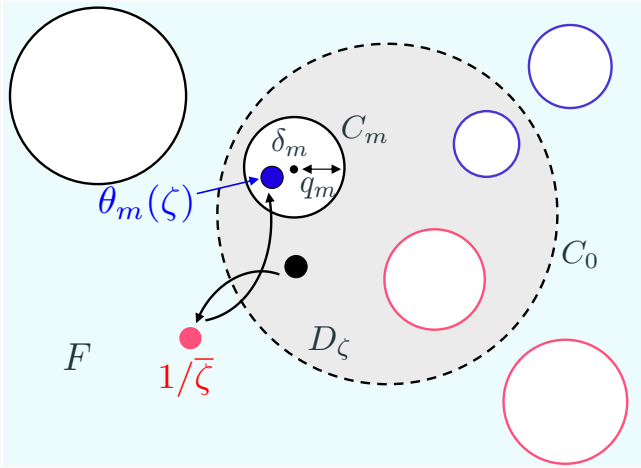
What carries over

An even number of inversions is holomorphic, and the sign in front of $\log$ is $+$. Those compositions are the factors of $P(\zeta, \rho)$.

One generator for each hole

An $(M+1)$-connected circular domain has hole centres δ_m and radii q_m. Inversion in circle m is

$$S_m(\zeta) = \overline{\delta_m} + \frac{q_m^2}{\zeta - \delta_m}.$$



That inversion, followed by inversion in the unit circle, is the Möbius map

$$\theta_m(\zeta) = \overline{S_m}(1/\zeta) = \delta_m + \frac{q_m^2 \zeta}{1 - \overline{\delta_m}\,\zeta}.$$

Inversion in C_m and then in C_n is $\theta_n \theta_m^{-1}$. $M = 2$ means three boundary circles.

The Schottky group

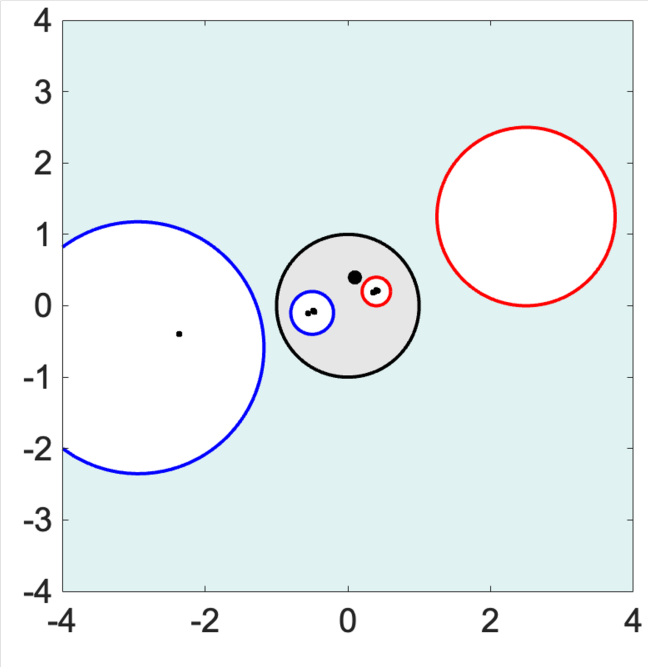
The $2M$ maps $\theta_m^{\pm 1}$ generate a free group Θ, the Schottky group of the circles:

$$\Theta = \{\theta_m, \theta_m^{-1} \mid m = 1, \ldots, M\}. \tag{3}$$

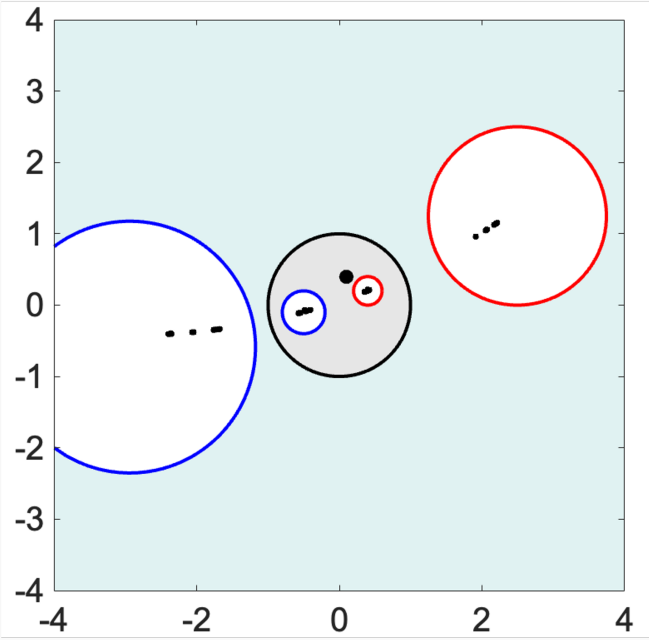
Every reflected copy of the vortex is $\theta(a)$ for some $\theta \in \Theta$.

Level 2

$\{\theta_1, \theta_2, \theta_1\theta_2, \theta_1^2, \theta_2^2, \theta_2\theta_1, \theta_1\theta_2^{-1}, \theta_1^{-1}\theta_2\}$



Level 6 (728 elements)



The prime function over the group

Drop the identity, and keep only one of θ and θ^{-1}. Call that set Θ'' (if $\theta_2\theta_3^{-1}$ is kept, $\theta_3\theta_2^{-1}$ is dropped).

Schottky–Klein prime functions (D. Crowdy, 2005)

$$\omega(\zeta, a) = (\zeta - a) \prod_{\theta \in \Theta''} \frac{(\theta(\zeta) - a)(\theta(a) - \zeta)}{(\theta(\zeta) - \zeta)(\theta(a) - a)}. \tag{4}$$

- The annulus is the special case. When $M = 1$, this product is Wednesday's $P(\zeta, \rho)$.
- This can be seen as a generalization of the annulus case.

Schottky–Klein prime functions (D. Crowdy, 2005)

$$\omega(\zeta, a) = (\zeta - a) \prod_{\theta \in \Theta''} \frac{(\theta(\zeta) - a)(\theta(a) - \zeta)}{(\theta(\zeta) - \zeta)(\theta(a) - a)}. \tag{5}$$

- There is a single zero at $\zeta = \theta(a)$.
- There also has a zero at $\zeta = \theta^{-1}(a)$ because it satisfies $\theta(\zeta) = a$.

The vortex on every boundary circle

$$w(\zeta) = \frac{\Gamma}{2\pi \mathrm{i}} \log\left(\frac{\omega(\zeta, \alpha)}{|\alpha|\, \omega(\zeta, 1/\overline{\alpha})} \right). \tag{6}$$

$\mathrm{Im}\, w$ is constant on each boundary circle, the same test as on the annulus.

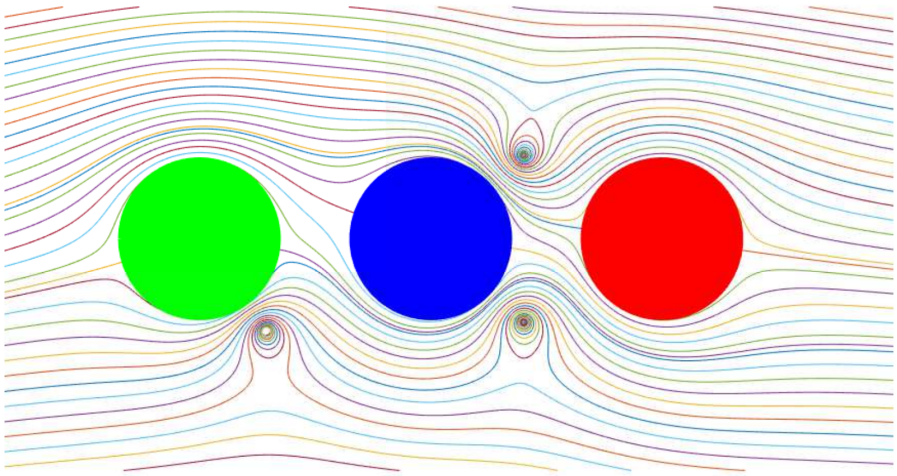


Figure 15.6. *Streamline distribution associated with uniform flow around three circular cylinders in the presence of three point vortices. There is no circulation around the cylinders. This figure was plotted using formulas (15.79)–(15.80).*

Crowdy, *Solving Problems in Multiply Connected Domains*, SIAM, 2020.

The product runs over words

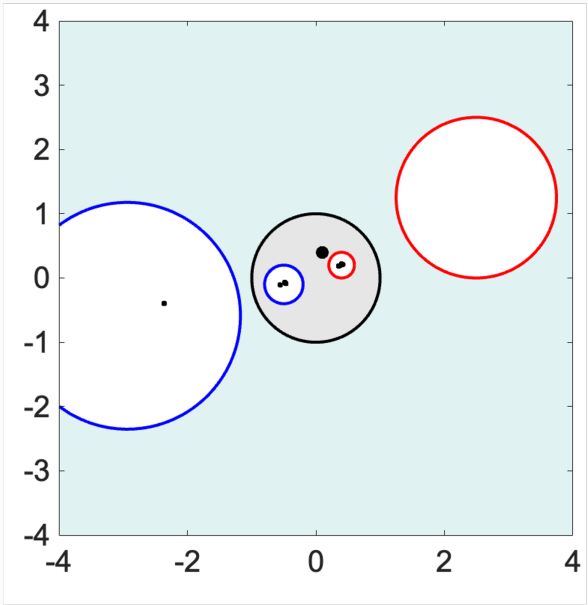
With M holes the generators do not commute. A reduced word of length L never cancels a generator against its own inverse, and there are

$$2M(2M-1)^{L-1} \tag{7}$$

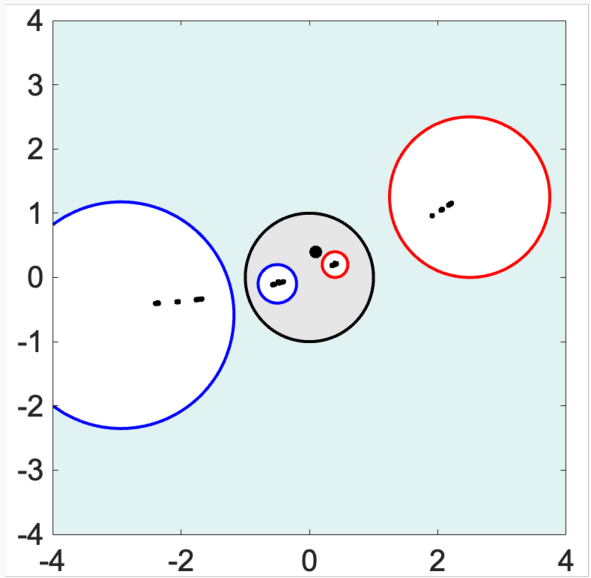
such words. The product for ω keeps one of θ and θ^{-1}, hence $M(2M-1)^{L-1}$ factors of length L. For two holes that sequence is $2, 6, 18, \ldots$, against two factors at each n on the annulus.

Level 2

$\{\theta_1, \theta_2, \theta_1\theta_2, \theta_1^2, \theta_2^2, \theta_2\theta_1, \theta_1\theta_2^{-1}, \theta_1^{-1}\theta_2\}$



Level 6 (728 elements)



One identity for each generator

Each generator acts on the prime function by a known factor (Crowdy, 2020):

$$\omega(\theta_m(\zeta), a) = \Gamma_m(\zeta, a)\,\omega(\zeta, a), \tag{8}$$

$$\Gamma_m(\zeta, a) = \sqrt{\theta_m'(\zeta)}\,\exp\Big(-2\pi\mathrm{i}\big(v_m(\zeta) - v_m(a) + \tfrac{1}{2}\tau_{mm}\big)\Big). \tag{9}$$

v_m is analytic off the holes and increases by 1 around the m-th hole. The period $\tau_{mm} = v_m(\theta_m(\zeta)) - v_m(\zeta)$ is a constant, fixed by the circles. The square root is the branch with $\omega(\zeta, a) \sim (\zeta - a)$.

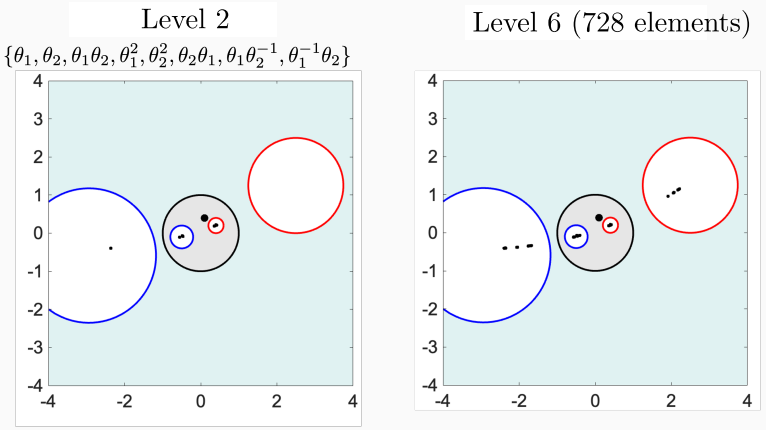
On the annulus both generators have $\Gamma_m = -\zeta^{-1}$. That pair is FE1 and FE2.

When the circles get close

Truncation is not free

On the annulus the factors are powers ρ^{2n}. If ρ is small, ρ^{2n} decays at once and a short product is enough. If $\rho \to 1$, the inner circle sits close to the outer one, and many n are needed before the truncated product is accurate.

With several holes the same fact is sharper. A gap between two circles is the local ρ. Words of length L contribute factors that shrink only when that gap is large.



Close circles mean a large product

- At word length L the product for ω already has $M(2M-1)^{L-1}$ factors.
- For $M = 2$: $2, 6, 18, 54, \ldots$.
- Circles that almost touch force a large L before the next images are negligible.
- A truncated product then leaves a visible residual on the identities, and on $\operatorname{Im} w$ on the boundary.

Numerical bottleneck

The classical construction still defines ω. It becomes expensive, and a short cut at $N_{\max}$ becomes inaccurate, precisely when the holes are close.

Bridge to a network

Product	Network
Infinite images, cut at $N_{\max}$	Constraints only: CR + identities
Accurate when gaps are wide	Same loss for every M
Slow when circles nearly touch	No enumeration of words in the loss

Idea

Keep the Schottky–Klein identities. Replace the truncated product by a network trained on those identities. The product remains the stored check, not the training data.

A PINN for the prime function

Today's question

The product defines P on the annulus, and ω for several holes. When the circles are close, that product is the object we trust least at a fixed $N_{\max}$.

Question

Can a network recover the same function from the **Cauchy–Riemann equations** and the **functional equations**, without being shown the values of P on the annulus?

The worked example is one hole: `03_schottky_klein_prime_nn`, with $\rho = 0.3$. Several holes use the same loss, with one residual per Schottky generator.

From Monday and Wednesday

Monday

A real network $u_\theta(x)$.
The loss was a differential equation and a boundary value.
A small loss was not always the exact solution.

Wednesday

$P(\zeta, \rho)$ is one holomorphic function on the annulus.
It is fixed by an infinite product, by inversion, and by quasi-periodicity.

Today the unknown is complex, so the network has two real outputs. The product plays the role that e^{-x} played on Monday: the check after training.

Representation

What the network outputs

Write $\zeta = x + \mathrm{i}y$ and $P = u + \mathrm{i}v$. `PrimeNet` is an ordinary MLP

$$(x, y) \longmapsto (u_\theta, v_\theta), \tag{10}$$

with $\tanh$ activations. In the notebook: width 80, depth 6.

Complex arithmetic ($1/\zeta$, products) is written by hand on these two outputs. Nothing in the architecture forces $u + \mathrm{i}v$ to be holomorphic.

Cauchy–Riemann as a loss

Holomorphicity is the pair

$$\partial_x u = \partial_y v, \qquad \partial_y u = -\partial_x v. \tag{11}$$

Both derivatives come from `autograd`, as on Day 1. At collocation points in the annulus,

$$\mathcal{L}_{\mathrm{CR}} = \mathrm{mean}\big((\partial_x u - \partial_y v)^2 + (\partial_y u + \partial_x v)^2\big). \tag{12}$$

What this term can do

It penalises a non-holomorphic output. It does not select the prime function: many holomorphic functions have $\mathcal{L}_{\mathrm{CR}} = 0$.

Identities

Two identities from Wednesday

On the annulus, $P(\cdot, \rho)$ satisfies

$$P(\zeta^{-1}) + \zeta^{-1} P(\zeta) = 0, \tag{13}$$

$$P(\rho^2 \zeta) + \zeta^{-1} P(\zeta) = 0. \tag{14}$$

Sample ζ in $\rho + 0.05 \le |\zeta| \le 0.95$ and penalise both residuals:

$$\mathcal{L}_{\mathrm{FE}} = \mathrm{mean}|P_\theta(\zeta^{-1}) + \zeta^{-1} P_\theta(\zeta)|^2 + \mathrm{mean}|P_\theta(\rho^2 \zeta) + \zeta^{-1} P_\theta(\zeta)|^2. \tag{15}$$

The network is evaluated at ζ, at $1/\zeta$, and at $\rho^2 \zeta$.

The zero function is a solution

Both identities are linear and homogeneous. If P satisfies them, then so do 0 and cP for any constant c.

Homogeneous constraints

$\mathcal{L}_{\mathrm{FE}} = \mathcal{L}_{\mathrm{CR}} = 0$ does not imply that the network has learned P. The zero network is a perfect minimiser of those two terms.

A third condition has to fix the scale. That condition is the only place where the classical product enters the loss.

Gauge near $\zeta = 1$

The product has a simple zero at $\zeta = 1$, and

$$\frac{P(\zeta, \rho)}{1 - \zeta} \to C = \prod_{n=1}^{\infty} (1 - \rho^{2n})^2 \qquad (\zeta \to 1). \tag{16}$$

For $\rho = 0.3$ the truncated product gives $C \approx 0.813$.

Two extra terms, sampled near the outer circle and at $\zeta = 1$:

$$P_\theta(\zeta) \approx (1 - \zeta)\, C, \qquad P_\theta(1) = 0. \tag{17}$$

What is, and is not, given to the network

The values of P on the annulus are **not** training data. The number C, and the condition $P(1) = 0$, are.

Check

How to read a small loss

Training never sees P_{ref} except through C. The relative error

$$\frac{\|P_\theta - P_{\mathrm{ref}}\|_{L^2(\mathrm{annulus})}}{\|P_{\mathrm{ref}}\|_{L^2(\mathrm{annulus})}} \tag{18}$$

is a diagnostic, as on Day 1.

A small $\mathcal{L}$ is not the answer

If the functional-equation residuals are tiny and this ratio is not, the constraints did not select P. The usual cause is a weak gauge, or a weight that lets one term dominate.

What is being compared

1. Truncate $P(\zeta, \rho)$ and form w. This is Wednesday's potential.
2. Insert the trained P_θ into the same formula.
3. Compare streamlines, and quote the relative L^2 error of P_θ against the product.

The same loss, one term per hole

The network is still $(x, y) \mapsto (u_\theta, v_\theta)$. Cauchy–Riemann does not change. The gauge is a simple zero at the chosen point a,

$$\omega_\theta(a, a) = 0, \qquad \frac{\omega_\theta(\zeta, a)}{\zeta - a} \to 1 \quad (\zeta \to a), \tag{19}$$

instead of $P(1) = 0$ and the constant C.

$\mathcal{L}_{\mathrm{FE}}$ is one residual for each generator: the mean of $|\omega_\theta(\theta_m(\zeta), a) - \Gamma_m(\zeta, a)\,\omega_\theta(\zeta, a)|^2$. For $M = 1$ these two residuals are FE1 and FE2. The reference after training is the product cut at word length $N_{\max}$.

What becomes harder

- **Where to sample.** Points stay inside the outer circle and outside every hole. A point in a hole does not test the identity.
- **The gauge.** The identities stay homogeneous, so $\omega \equiv 0$ is still a minimiser until the simple zero at a is weighted strongly enough. More holes add more terms that can dominate that weight.

Not in the notebook yet

`03_schottky_klein_prime_nn` trains $M = 1$ only. Several holes are the same program with a list of θ_m. That is the regime where the truncated product is most expensive, and where a PINN is most worth trying.

Application
Cell culture experiments and multiply connected domains

Why look at a domain with holes

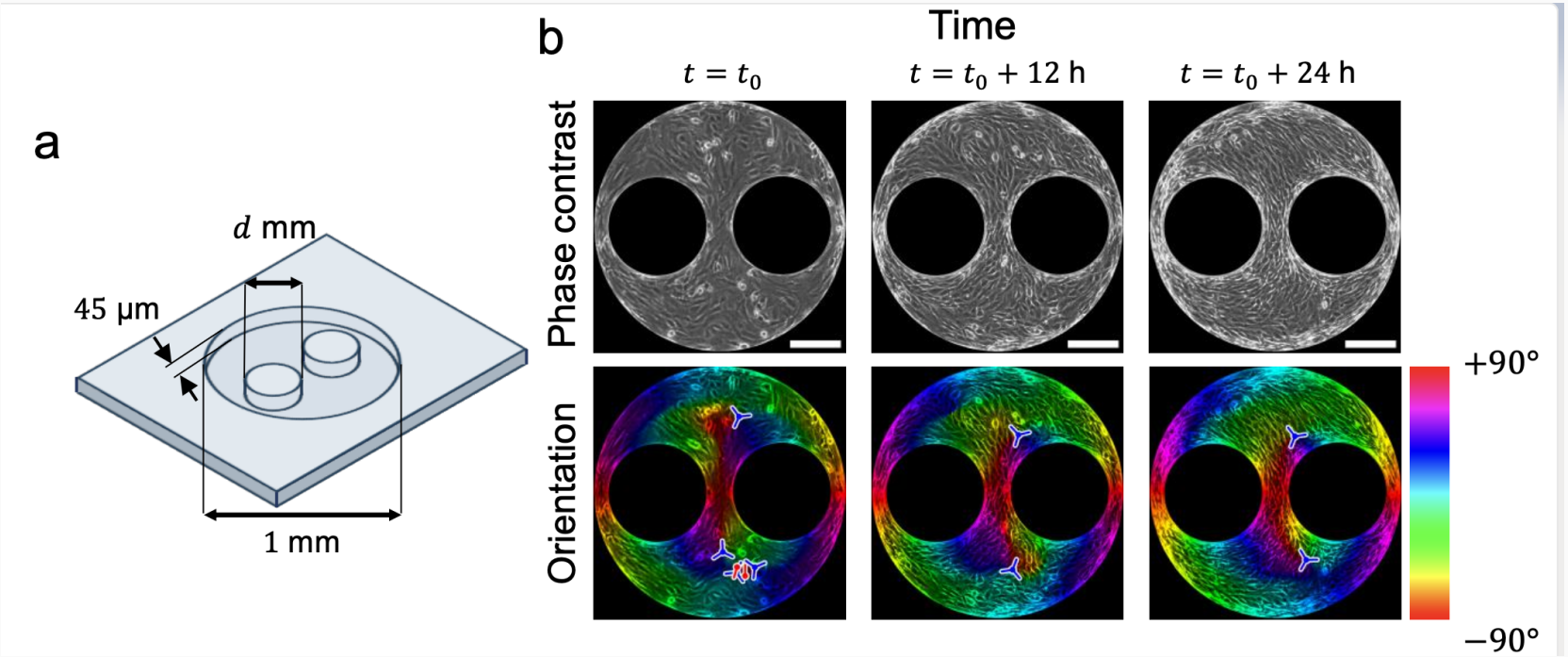
- A disk ($\chi = +1$) stably holds two defects of charge $+1/2$.
- An annulus ($\chi = 0$) may hold one $+1/2$ and one $-1/2$, or none. The inner radius decides.
- Tissue often wraps two or more obstacles. Then $\chi = 1 - h < 0$, so the total charge is negative.
- A short-axis slice of heart muscle is one such domain: two $-1/2$ defects sit where the ventricles meet.

The question

Does the size of those obstacles choose where the negative defects sit, and so the symmetry of the alignment?

Two holes fix the total charge

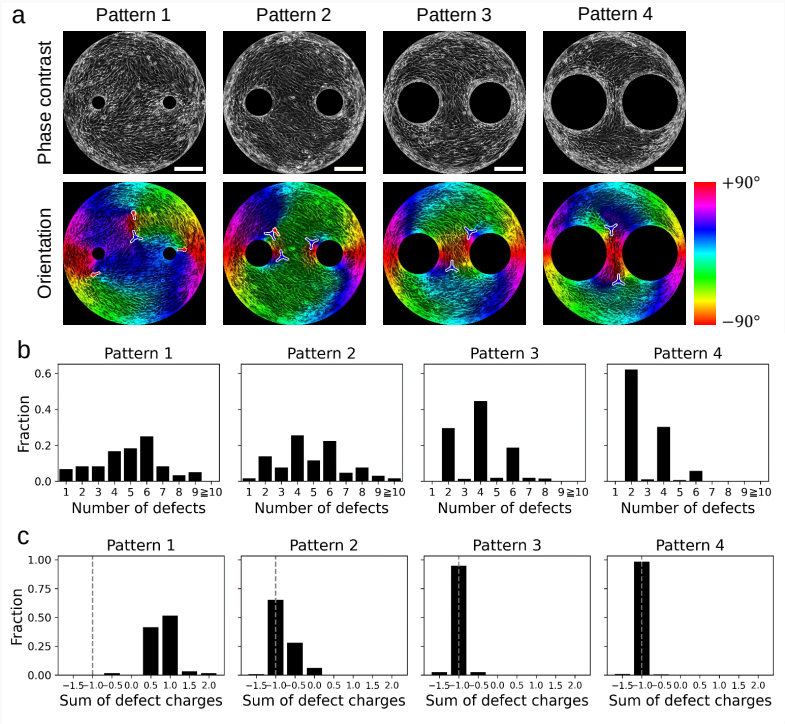
- A $1\,\mathrm{mm}$ well with two pillars: triply connected, $\chi = 1 - h = -1$.
- Tangential anchoring forces the total defect charge to equal χ.
- Pairs annihilate. Two defects of charge $-1/2$ remain, and the mean charge stays -1.



Kamao et al., arXiv:2609.03328, Fig. 1.

Large obstacles leave charge -1

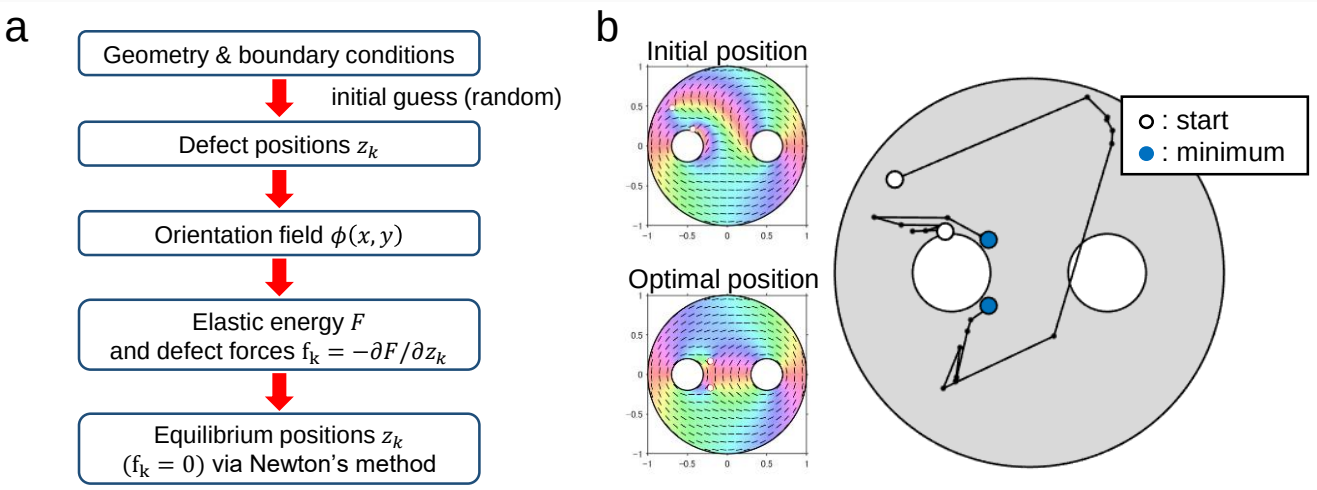
- Pillar diameter $d = 0.1, 0.2, 0.3, 0.4\,\mathrm{mm}$ (Patterns 1–4).
- For the two largest, more than 95% of wells have total charge -1.
- For the two smallest, anchoring on the pillars fails.



The orientation is a prime function

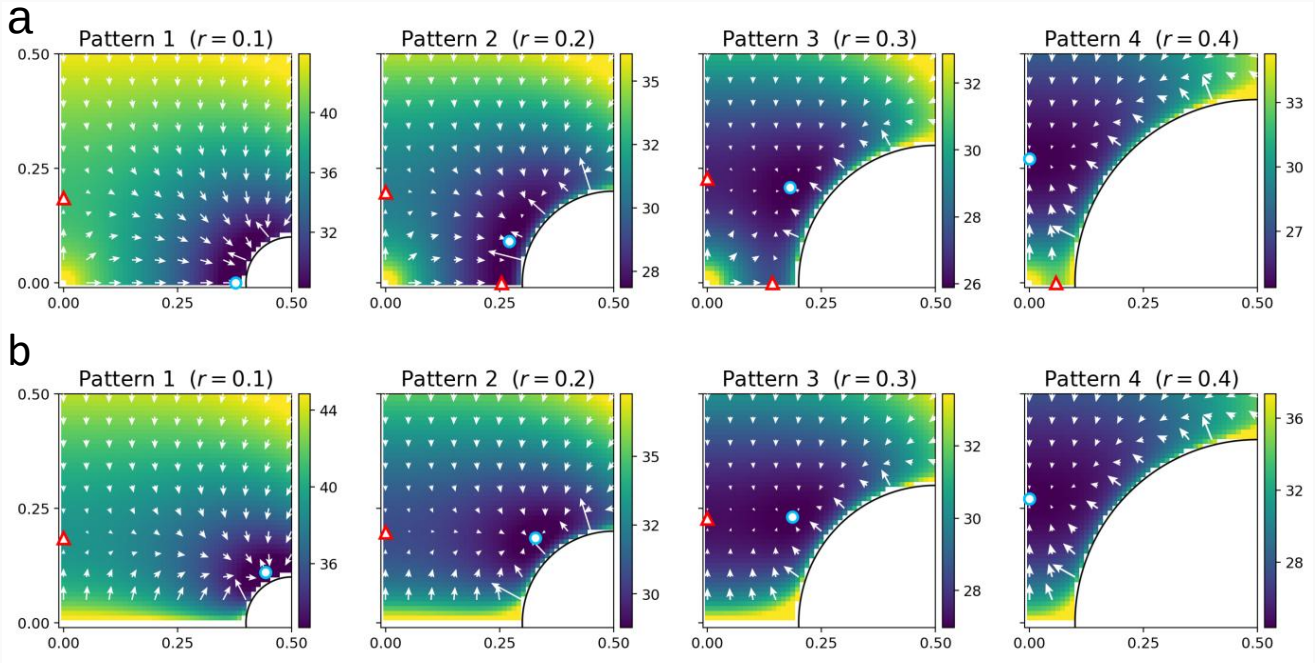
Two $-1/2$ defects at z_1, z_2. The angle ϕ is harmonic off the defects, winds by $-\pi$ about each, and is tangential on every circle.

$$\phi = \operatorname{Re} f, \qquad F = \frac{K}{2} \iint_{\tilde{D}} \left|\partial_z f\right|^2 \mathrm{d}x\, \mathrm{d}y. \tag{20}$$



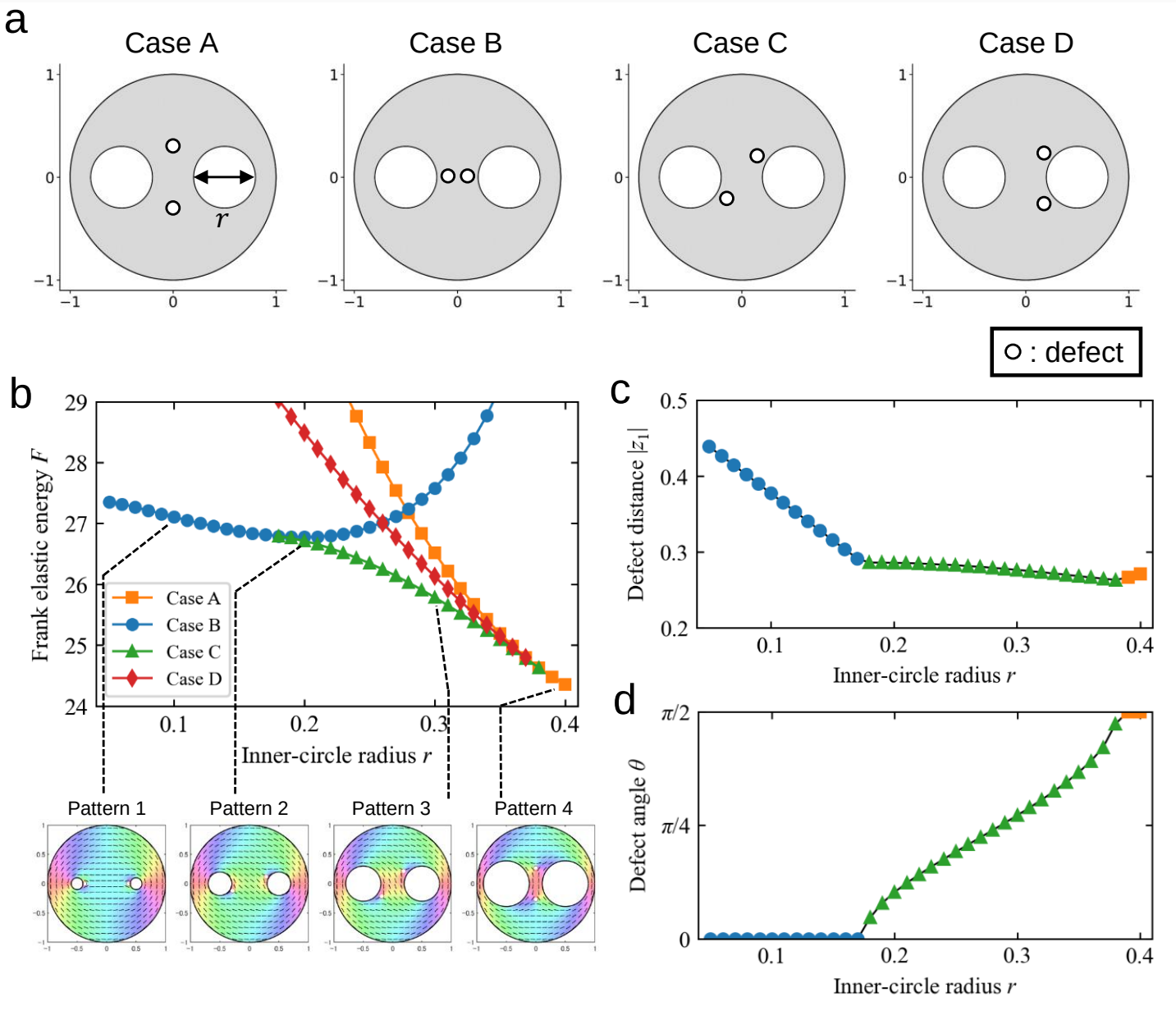
The minimizer leaves the axis

- Symmetric pairs: $z_2 = -z_1$, or $z_2 = \overline{z_1}$.
- Blue: minima of F. Red: saddles.
- As the inner radius grows, the minimum moves off the horizontal axis, then onto the vertical axis.



Same paper, Fig. 7. Arrows: force on the first defect.

The radius selects the symmetry



Close

Thursday, in one pass

Annulus: P is a truncated product. Several holes: ω over Schottky words. Close circles make that truncation expensive. A PINN keeps the same identities as a loss; the product is the stored check. Two holes in a cell monolayer: the prime function builds the angle,
and the obstacle radius selects the symmetry.

Exercise

Apply the stored run of `03_schottky_klein_prime_nn.ipynb` with $\rho = 0.3$.

1. Record the relative L^2 error against `P_ref`, and the separate pieces of $\mathcal{L}$ at the last epoch.
2. Say whether a small functional-equation loss, with the gauge removed, still produced P.
3. Name which object in the vortex formula is replaced by the network.

References

- `work/pinn_learning/03_schottky_klein_prime_nn.ipynb`
- Crowdy (2020), *Solving Problems in Multiply Connected Domains*, SIAM. The factor Γ_m for each Schottky generator.
- Raissi, Perdikaris, Karniadakis (2019), for the Monday residual loss
- Cuomo et al. (2022), *J. Sci. Comput.* **92**, 88: the loss, the initialization, and the open convergence question
- Farea, Yli-Harja, Emmert-Streib (2024), *AI* **5**(3), 1534–1557: a boundary constraint, checked by a test
- Kamao, Miyoshi, Nara, Miyazako (2026), arXiv:2609.03328. Two $-1/2$ defects in a triply connected cell monolayer.